

Student Information
Management System
Requirement Specification

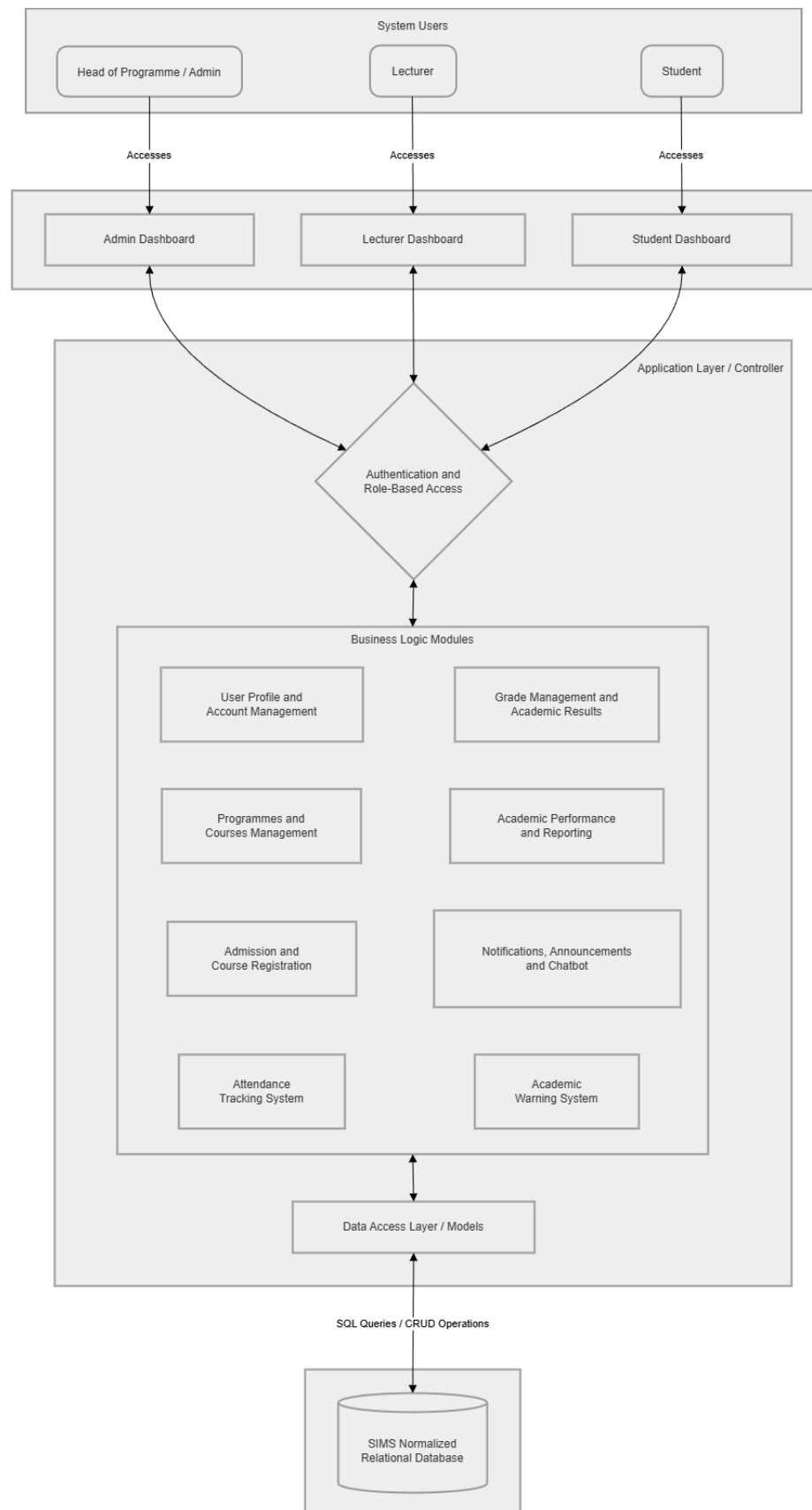
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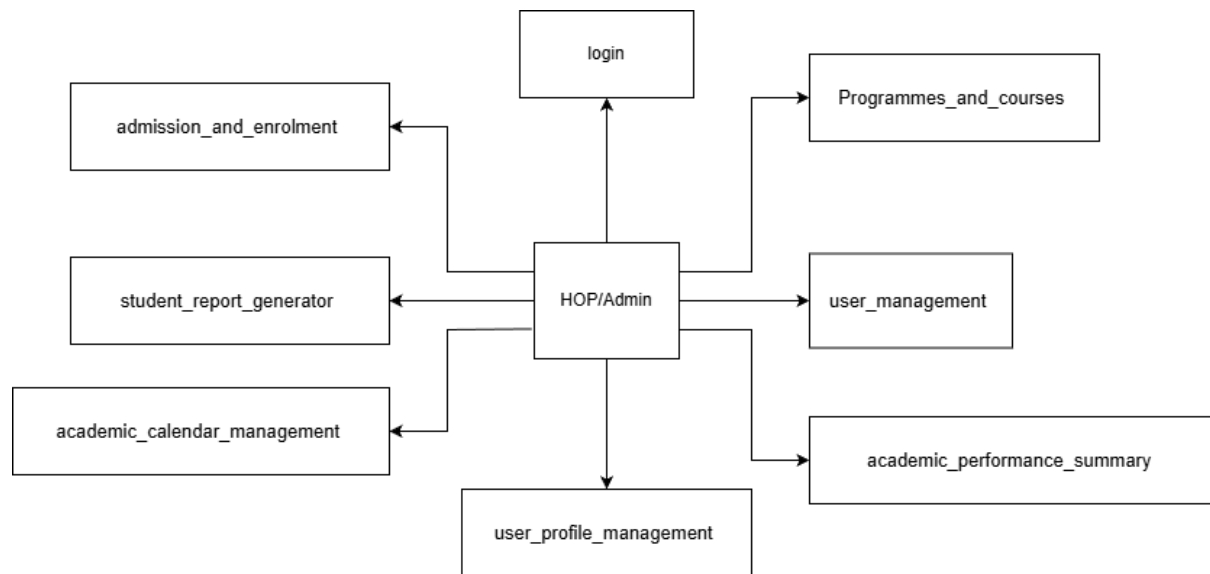
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1.0 System Architecture



2.0 System Requirements Specification

2.1 Head of Programme (Admin) Module



2.1.1 User management

1. The system shall allow the HOP to create new user accounts for lecturers and students.
2. The system shall store and maintain basic user information, including but not limited to:
 - Full name
 - Email address
 - Student ID (Auto Assigning; only for student)
 - Role (Lecturer / Student)
3. The system shall support role-based access control (RBAC), where access permissions are assigned according to user roles.

2.1.2 Admission and enrolment

1. The system shall allow HOP to approve or reject student admission applications.
2. Handling add/drop course requests
3. Registering students into programmes or courses
4. Academic status (active, deferred, graduated, withdrawn)

2.1.3 Academic performance summary

1. The system shall display total student counts (active, graduated, dropped out).
2. The system shall show enrolment numbers by programme, course, and semester.
3. The system shall provide attendance rate summaries.
4. The system shall display student academic results (GPA/CGPA).
5. The system shall generate pass/fail rate statistics.
6. The system shall identify top-performing and at-risk students.
7. The system shall provide dashboards with key academic metrics.
8. The system shall present performance trends using charts and graphs.

2.1.4 Student Report Generator

1. The system shall generate reports in PDF and Excel formats, including graphs and charts, based on academic performance data.
2. The system shall allow generation of overall summary reports with filtering by all students or by specific courses.
3. The system shall support generation of individual student-specific academic reports.

2.1.5 Academic Calendar Management

1. The system shall allow the HOP to create and maintain semester schedules.
2. The system shall support setting and managing key academic dates, including:
 - Semester start and end dates
 - Examination periods
 - Registration deadlines
 - Holidays and academic breaks
3. The system shall allow updates and modifications when academic schedules change.
4. The system shall provide announcements management for academic-related updates.
5. The system shall send notifications (email and in-app) to students and lecturers for calendar updates and announcements.

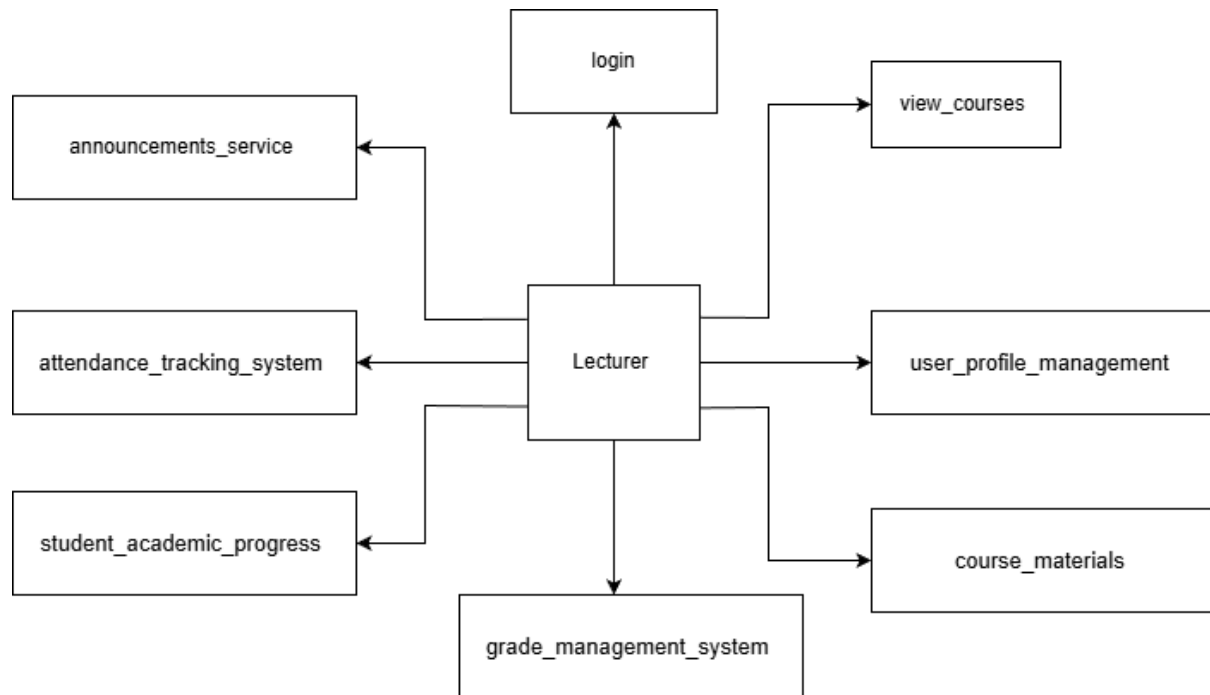
2.1.6 Programmes and Courses

1. The system shall allow the HOP to create and maintain study programmes (e.g., Diploma in IT, Degree in Business).
2. The system shall allow the HOP to create and maintain course information, including:
 - Course code
 - Course title
 - Credit value
 - Assigned lecturer
3. The system shall support assigning courses to specific programmes and semesters.
4. Auto assign to lecturer (optional)

2.1.7 User Profile Management

1. The system shall allow users to manage their personal details, including:
 - Name
 - Profile picture
 - Email address
2. The system shall allow users to update their profile information where permitted by their role.
3. The system shall support password management with the following rules:
 - Users can reset their password via email verification if forgotten
 - Users can change their password within the system by entering the current password for verification

2.2 Lecturer Module



2.2.1 View Courses

1. The system shall display a list of courses assigned to each lecturer.
2. The system shall provide detailed information for each course, including:
 - Course name and code
 - Semester or academic session
 - Schedule or class group
3. The system shall allow lecturers to view the list of students enrolled in each assigned course.
4. The system shall display student details, including:
 - Student name
 - Student ID
5. The system shall ensure that lecturers can only access information related to their assigned courses.

2.2.2 Attendance Tracking System

1. The system shall allow lecturers to record student attendance by marking status as:
 - Present
 - Absent
2. The system shall allow lecturers to update or correct attendance records when necessary.
3. The system shall allow modification of attendance status when a student provides valid justification (e.g., medical certificate, approved leave).
4. The system shall support QR code-based attendance, where students can scan a QR code to register their attendance.
5. The system shall ensure attendance records are updated in real time and stored securely for reporting purposes.

2.2.3 Grade Management System

1. The system shall allow lecturers to enter student marks for assessments, including:
 - Exams
 - Assignments
 - Quizzes
 - Projects
2. The system shall allow lecturers to update or correct entered marks and grades.
3. The system shall support grade modifications due to valid reasons such as remarking or late submissions.
4. The system shall allow lecturers to publish grades to students.
5. The system shall ensure that published grades are made visible to the respective students securely.

2.2.4 Student Academic Progress

Note: Requirement not clear

2.2.5 Announcements Service

1. The system shall allow lecturers to send announcements to students enrolled in a specific course.
2. The system shall ensure that announcements are delivered only to the selected course group.
3. The system shall display announcements to students within the system interface.
4. The system shall store announcement history for reference purposes.

2.2.6 Course Materials

1. The system shall allow lecturers to upload course learning resources, including:
 - Lecture notes (PDF, slides)
 - Reading materials
 - Assignments
2. The system shall associate uploaded materials with the relevant course.
3. The system shall allow students enrolled in the course to view and download the materials.
4. The system shall ensure that course materials are securely stored and access-controlled based on course enrollment.

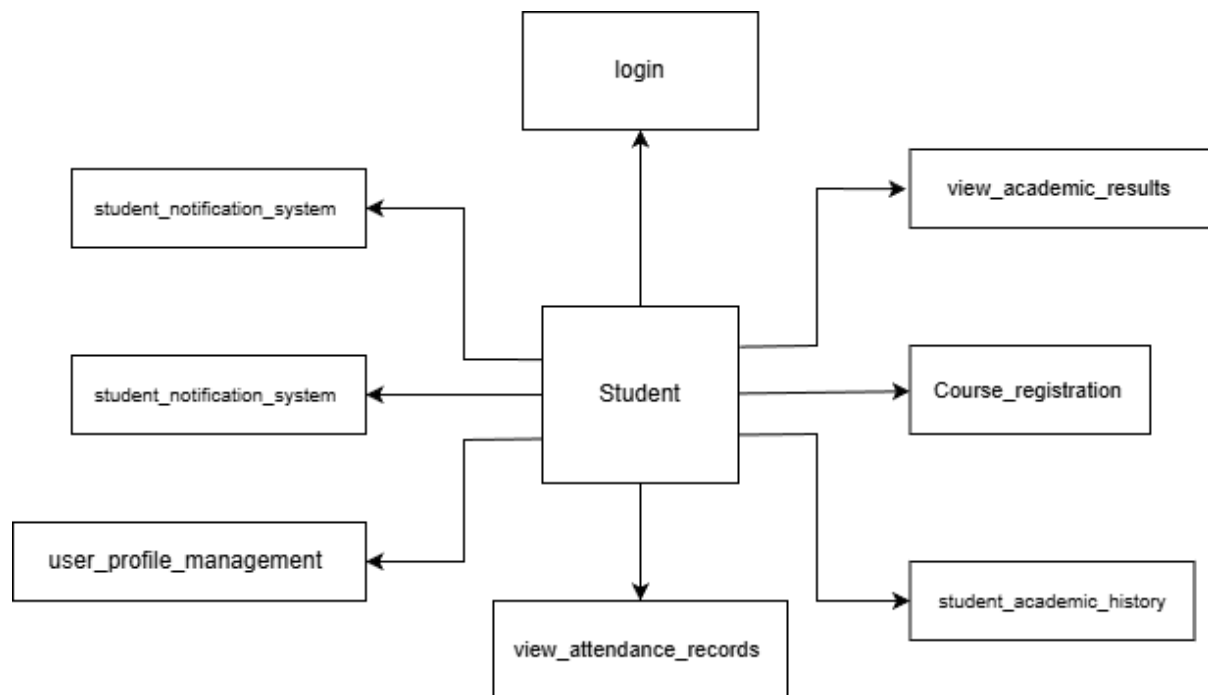
2.2.7 Identify students with poor attendance or academic performance

1. not clear with requirement

2.2.8 User Profile Management

4. The system shall allow users to manage their personal details, including:
 - Name
 - Profile picture
 - Email address
5. The system shall allow users to update their profile information where permitted by their role.
6. The system shall support password management with the following rules:
 - Users can reset their password via email verification if forgotten
 - Users can change their password within the system by entering the current password for verification

2.3 Student Module



2.3.1 Course Registration

1. The system shall allow students to register for selected courses for each semester.
2. The system shall validate course registration by checking:
 - Course availability
 - Credit limit restrictions
3. The system shall allow students to drop previously registered courses.
4. Automated course enrolment validation.
5. The system shall restrict course registration activities to permitted periods defined in the academic calendar, including:
 - Registration period (before semester start)
 - Add/drop period (early semester adjustment phase)

2.3.2 View Attendance Records

1. The system shall allow students to view their attendance records for each course.
2. The system shall display attendance status (e.g., present, absent) in a structured format.
3. The system shall provide an attendance summary report for each course or semester.
4. The system shall ensure that attendance data is accessible only to the respective student.

2.3.3 View Academic Results

1. The system shall allow students to view results for specific courses, including:
 - Assignment name
 - Submission time
 - Submission status (submitted / missing)
 - Score
2. The system shall allow students to export results as PDF or Excel files.
3. The system shall provide an overall academic summary view, including:
 - Course code and name
 - Course grade
 - GPA / CGPA
4. The system shall support transcript generation for academic records.

2.3.4 Student Academic History

1. The system shall store all past academic records for each student.
2. The system shall maintain records of courses taken in each semester.
3. The system shall store grades received for each completed course.
4. The system shall record credits earned per course and semester.
5. The system shall ensure academic history is securely stored and accessible only to the respective student and authorized users.
6. Performance reports (not clear)

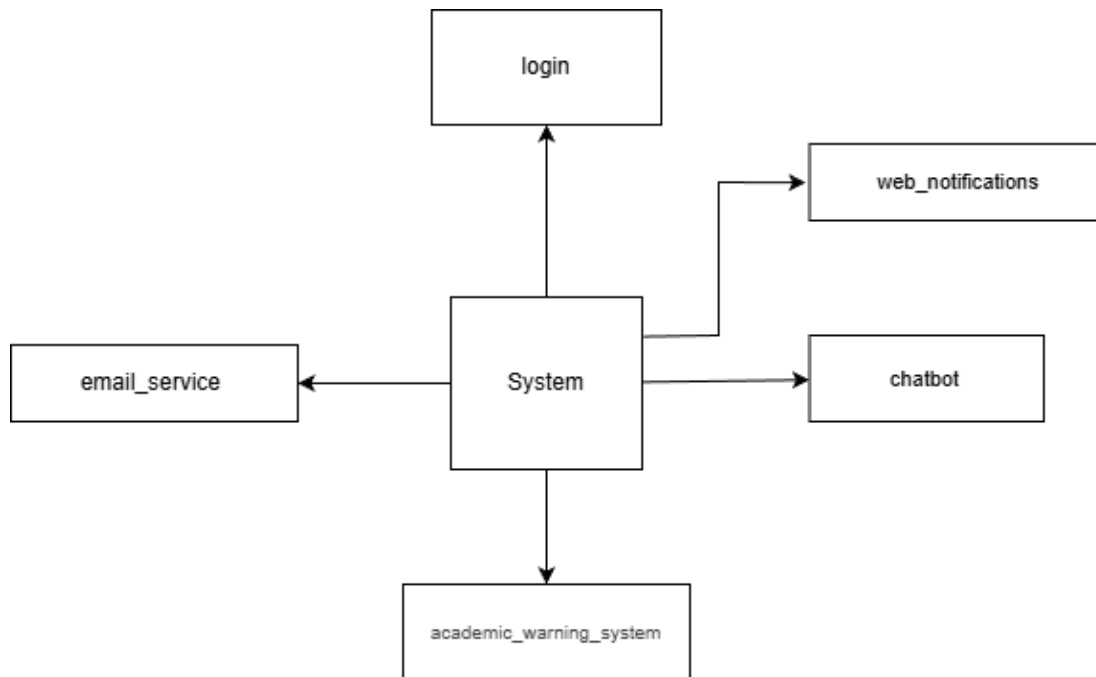
2.3.5 Student Notification System

1. The system shall deliver real-time notifications to students.
2. The system shall send notifications related to:
 - Lecturer announcements
 - Grade updates
 - Assignment deadlines
 - Course changes
 - Timetable updates
 - Registration status updates
3. The system shall display notifications within the application interface.
4. The system shall ensure notifications are relevant to the respective student only.
5. The system shall maintain a history of notifications for reference purposes.

2.3.6 User Profile Management

7. The system shall allow users to manage their personal details, including:
 - Name
 - Profile picture
 - Email address
8. The system shall allow users to update their profile information where permitted by their role.
9. The system shall support password management with the following rules:
 - Users can reset their password via email verification if forgotten
 - Users can change their password within the system by entering the current password for verification

2.4 System module



The system shall provide core system-wide modules to support authentication, monitoring, communication, and user assistance.

2.4.1 Login

1. The system shall provide a login module for all users.
2. The system shall authenticate users based on credentials (e.g., username/email and password).
3. The system shall enforce role-based access control (HOP, Lecturer, Student).
4. The system shall direct users to different dashboards based on their assigned roles.

2.4.2 Academic Warning System

1. The system shall monitor student attendance and academic performance.
2. The system shall automatically notify students when attendance falls below a defined threshold.
3. The system shall detect frequent absences and trigger warning alerts.
4. The system shall issue academic warning notifications to students, and optionally to lecturers and administrators.
5. The system shall support an early warning (at-risk detection) mechanism for identifying underperforming students.

2.4.3 Email Service

1. The system shall support sending email notifications for academic activities.

2.4.4 Web Notifications

1. The system shall provide real-time in-app (web) notifications.
2. The system shall notify users of updates such as grades, announcements, deadlines, and course changes.

2.4.5 Chatbot

1. The system shall provide a chatbot accessible to all user roles (HOP, Lecturer, Student).